Math 6601 Homework 3

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Problem 1.

(a) Let $x = (2, -3, 0, 1)^T$. Then

$$\|x\|_1 = |2| + |-3| + |0| + |1| = 6,$$
$$\|x\|_2 = \sqrt{(2)^2 + (-3)^2 + (0)^2 + (1)^2} = \sqrt{14},$$

and

$$\|x\|_\infty = \max\{|2|, |-3|, |0|, |1|\} = 3.$$

(b) Let

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & -1 \\ 2 & 3 & 2 \end{bmatrix}.$$

If $A = \{a_{ij}\}$, then

$$\begin{aligned}\|A\|_1 &= \max_{j\in\{1,2,3\}} \sum_{i=1}^{3} |a_{ij}| = \max\{|2| + |4| + |2|, |-1| + |2| + |3|, |3| + |-1| + |2|\} \\ &= \max\{8, 6, 6\} = 8,\end{aligned}$$

and

$$\begin{aligned}\|A\|_\infty &= \max_{i\in\{1,2,3\}} \sum_{j=1}^{3} |a_{ij}| = \max\{|2| + |-1| + |3|, |4| + |2| + |-1|, |2| + |3| + |2|\} \\ &= \max\{6, 7, 7\} = 7.\end{aligned}$$

Problem 2.

(a) Let

$$A = \begin{bmatrix} 0 & -2 \\ -\frac{1}{8} & 0 \end{bmatrix}.$$

Then the characteristic equation of A is $\lambda^2 - \frac{1}{4} = 0$, so the eigenvalues of A are the roots $\lambda_1 = \frac{1}{2}$, and $\lambda_2 = -\frac{1}{2}$. Thus, $\rho(A) = \frac{1}{2} < 1$ implies that $\lim_{k\to\infty} A^k = 0$.

(b) Let

$$A = \begin{bmatrix} 1 & -1 \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}.$$

Then the characteristic equation of A is $\lambda^2 - \frac{5}{2}\lambda + 1 = 0$, or equivalently $(2\lambda - 1)(\lambda - 2) = 0$. Then the eigenvalues of A are the roots $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = 2$. Thus, $\rho(A) = 2 > 1$ implies that $\lim_{k\to\infty} A^k \neq 0$.

Problem 3.

Let

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 3 & 9 \\ 3 & 3 & 5 \end{bmatrix}.$$

To compute the condition number $\text{cond}(A)$ using $\|\cdot\|_2$, we use the fact that $\text{cond}(A) = \|A\|_2\|A^{-1}\|_2$. We know that $\|A\|_2 = \sqrt{\rho(A^TA)}$, and $\|A^{-1}\|_2 = \sqrt{\rho\left(\left(A^{-1}\right)^T A^{-1}\right)}$. We can compute

$$A^TA = \begin{bmatrix} 22 & 16 & 44 \\ 16 & 19 & 41 \\ 44 & 41 & 107 \end{bmatrix},$$

and

$$A^{-1} = \frac{1}{36}\begin{bmatrix} 12 & -8 & 12 \\ -12 & -7 & 15 \\ 0 & 9 & -9 \end{bmatrix},$$

and

$$\left(A^{-1}\right)^T A^{-1} = \frac{1}{648}\begin{bmatrix} 144 & -6 & -18 \\ -6 & 97 & -141 \\ -18 & -141 & 225 \end{bmatrix}.$$

The characteristic equation of A^TA is then $\det(A^TA - \lambda I) = -\lambda^3 + 148\lambda^2 - 932\lambda + 1296$. The exact value of the largest root of this equation is very complicated; instead, we can find an approximation of the largest root $\rho(A^TA) = \lambda_1 \approx 141.47711088$ using Newton's method with a very large initial guess.

Similarly, the characteristic equation of $\left(A^{-1}\right)^T A^{-1}$ is

$$\det\left(\left(A^{-1}\right)^T A^{-1} - \lambda I\right) = -\lambda^3 + \frac{466}{648}\lambda^2 - \frac{47952}{648^2}\lambda + \frac{209952}{648^3} = 0,$$

which we can also solve approximately with Newton's method using a large initial guess to get an approximation of the largest root $\rho\left(\left(A^{-1}\right)^T A^{-1}\right) = \lambda_1 \approx 0.48868217$.

Then the condition number of A using $\|\cdot\|_2$ is

$$\text{cond}(A) = \|A\|_2\|A^{-1}\|_2 = \sqrt{\rho(A^TA)\rho\left(\left(A^{-1}\right)^T A^{-1}\right)} \approx \sqrt{141.47711088 \cdot 0.48868217} \approx 8.314887.$$

Problem 4.

Let A be a real, symmetric, positive definite matrix. Then A^{-1} is also a symmetric, positive definite matrix.

Proof. First, we show that A^{-1} is symmetric. This is the case because

$$\begin{aligned}\left(A^{-1}\right)^T A &= \left(A^T A^{-1}\right)^T = \left(AA^{-1}\right)^T = I^T = I,\\ A\left(A^{-1}\right)^T &= \left(A^{-1}A^T\right)^T = \left(A^{-1}A\right)^T = I^T = I,\end{aligned}$$

by the symmetry of A, which implies that $\left(A^{-1}\right)^T$ is the inverse of A. That is, $A^{-1} = \left(A^{-1}\right)^T$, or A^{-1} is symmetric.

Second, we show that A^{-1} is positive definite. This is the case because, for any vector $x \neq 0$, we can set $y = A^{-1}x$, so that $x = Ay$, giving

$$x^T A^{-1} x = y^T A^T A^{-1} A y = y^T A A^{-1} A y = y^T A y > 0$$

by the positive-definiteness of A. Thus, A^{-1} is positive definite. $\square$